

Islamic University of Technology

Board Bazar, Gazipur, Dhaka



SWE 4404 – Software Project Lab-II

Software Design Report

ALUMNET

Student-Alumni Mentorship and Career Guidance Platform

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SPL Team - 11

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Chapter 1

Project Overview

Project Name: Alumnet

This project is a web-based Student–Alumni Mentorship and Career Guidance platform designed to connect students with alumni for mentorship, networking, and career development. It provides an organized system for students to discover and connect with relevant alumni based on shared interests, career goals, and professional expertise. Through mentorship and referral features, students can seek guidance, career advice, and professional opportunities, while alumni can support students by sharing experiences, posting job or internship opportunities, and offering referrals. Additionally, the platform fosters a collaborative community through discussion forums, knowledge sharing, and event updates, creating a centralized ecosystem that supports student growth, alumni engagement, and career readiness.

Chapter 2

Software Requirements & Specifications:

[Group-11 \[103, 137, 156\] Summer 24-25 SPL-II Requirement Collection Document](#)

Chapter 3

Key Features

3.1 User Authentication & Authorization

The platform will provide a secure authentication system for both students and alumni. Users will be able to register, log in, and manage their accounts safely using Firebase Authentication. Role-based authorization will ensure that users can only access features and actions permitted for their account type.

Use Case: A student or alumnus registers on the platform, logs in securely, and accesses features based on their assigned role.

3.2 User Profile System

Each user will have a personalized profile containing information such as academic background, skills, career interests, experience, and social links. Students can showcase their goals and interests, while alumni can share their professional journey and expertise to help students connect with suitable mentors.

Use Case: A student creates a profile with skills and career goals, while an alumni member updates professional information to help students connect with suitable mentors.

3.3 Alumni Search & Filtering

Students will be able to search and filter alumni based on different criteria such as department, company, job role, skills, and areas of expertise. This feature will help students quickly find alumni who match their career interests and mentorship needs.

Use Case: A student searches for alumni working in a specific company or field to receive career guidance and mentorship.

3.4 Attribute-Based Mentor Matching

The platform will suggest suitable mentors to students based on shared attributes such as career goals, skills, interests, and preferred fields. This matching system will make it easier for students to discover alumni who can provide relevant guidance and support.

Use Case: The platform recommends alumni mentors to students based on shared interests, skills, and career aspirations.

3.5 Mentorship & Referral System

Students can send mentorship or referral requests to alumni through the platform. Alumni will be able to accept or reject requests and manage ongoing mentorship connections. This feature will help students receive career advice, interview preparation support, and professional referrals. Mentors will be able to create customized task lists and learning goals for mentees after mentorship sessions, allowing structured guidance and progress tracking throughout the mentorship journey.

Use Case: A student sends a mentorship request to an alumni mentor, receives guidance, and follows a personalized roadmap or task list created by the mentor.

3.6 Post & Feed System

Users will be able to create and share posts related to career advice, achievements, opportunities, and experiences. A centralized feed system will allow students and alumni to stay updated with community activities and shared knowledge.

Use Case: Alumni share internship opportunities or career advice through posts, while students interact through likes and comments.

3.7 Real-Time Chat System

The platform will include a real-time chat system that allows direct communication between students and alumni. Users can discuss

mentorship topics, career guidance, and professional opportunities instantly through one-to-one messaging.

Use Case: A student communicates directly with an alumni mentor to discuss interview preparation, career planning, or project guidance.

3.8 Community & Events

The platform will support community engagement through discussion spaces and event updates. Users can participate in discussions, share knowledge, and stay informed about seminars, workshops, networking sessions, and other career-related events.

Use Case: Users participate in discussion forums and register for workshops, seminars, and networking events hosted through the platform.

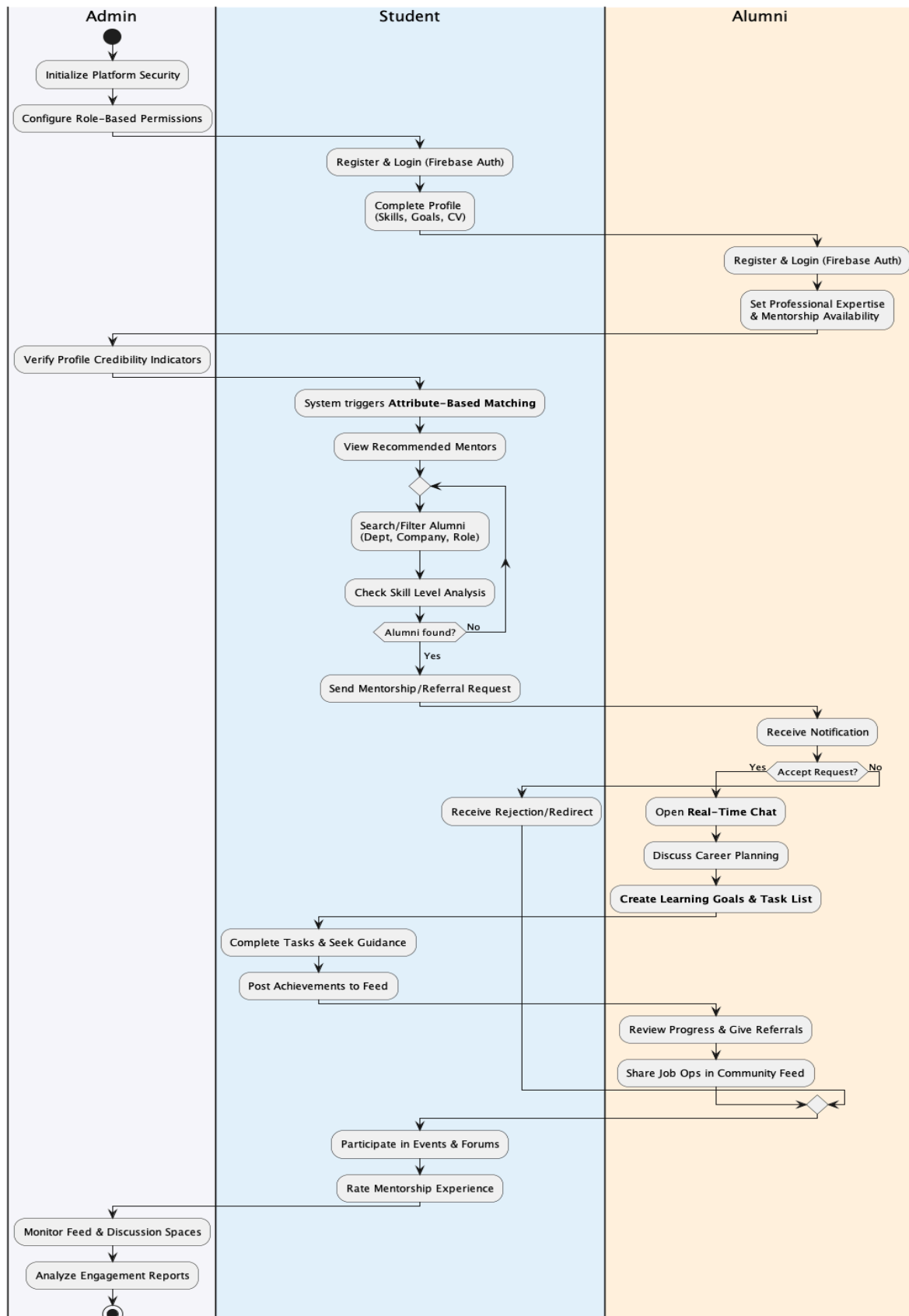
3.9 Additional Features

- **Skill Level Analysis:** User skills can be analyzed based on profile information, projects, certifications, and experience to better represent expertise and career readiness
- **Profile Credibility Indicators:** Verified achievements and contributions can be highlighted to help students identify reliable mentors and professionals

Use Case: The platform highlights verified achievements and analyzes user skills to improve mentor credibility and student career readiness.

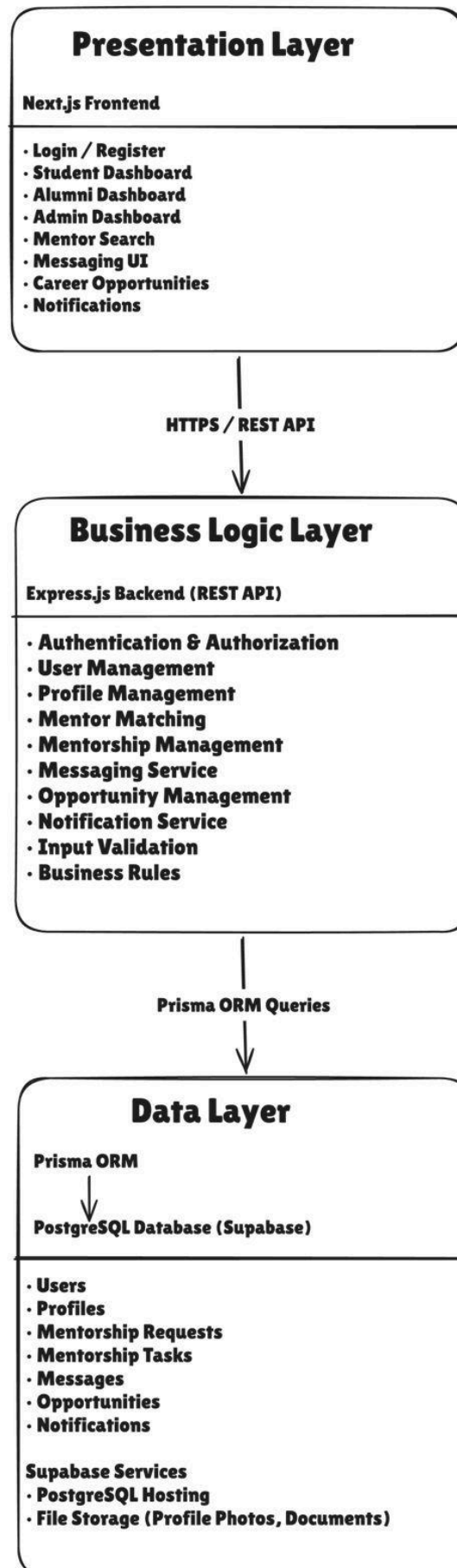
Chapter 4

Flow Chart



Chapter 5

System Architecture:



Architecture Explanation

Presentation Layer:

The Presentation Layer is developed using Next.js and serves as the user interface of the ALUMNET platform. It allows students, alumni, and administrators to interact with the system through responsive web pages. This layer is responsible only for displaying data and collecting user input. It communicates with the backend through secure REST API calls over HTTPS.

Main responsibilities include:

1. User Authentication Interface
2. Student Dashboard
3. Alumni Dashboard
4. Admin Dashboard
5. Mentor Search
6. Messaging Interface
7. Career Opportunity Portal
8. Notification Display

Business Logic Layer:

The Business Logic Layer is implemented using Express.js. It processes incoming requests from the frontend, enforces business rules, validates user input, performs authentication and authorization, and coordinates communication with the database through Prisma ORM.

Major services include:

1. Authentication & Authorization
2. User Management
3. Profile Management
4. Mentor Discovery
5. Mentorship Request Handling
6. Messaging Service
7. Opportunity Management

8. Notification Service

This layer ensures that all application logic remains separate from the user interface and database.

Data Layer:

The Data Layer is implemented using PostgreSQL, hosted on Supabase, with Prisma ORM acting as the database abstraction layer.

Prisma simplifies database operations by mapping database tables to application models and generating type-safe queries.

The database stores:

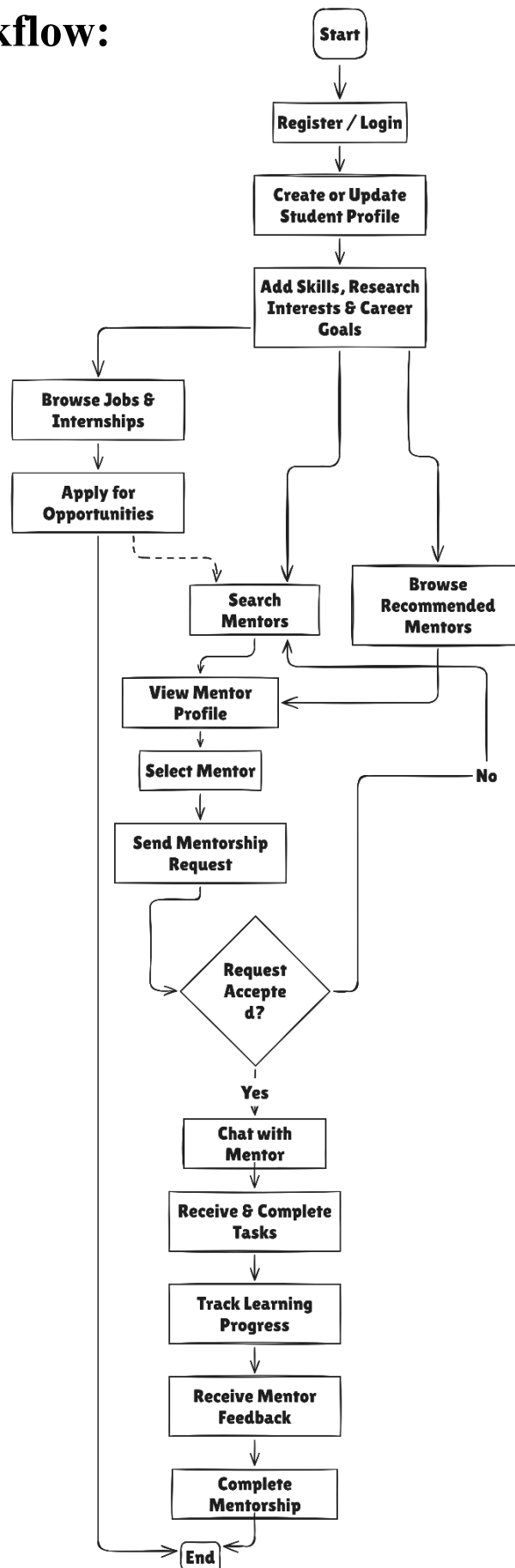
1. User Information
2. Profiles
3. Mentorship Requests
4. Mentorship Tasks
5. Messages
6. Career Opportunities
7. Notifications

Supabase additionally provides:

1. Managed PostgreSQL Hosting
2. Object Storage (for profile images and documents)

Chapter 6

Students Workflow:



Step 1: Register / Login

The workflow starts when a student creates a new account or logs into an existing account. Authentication allows the student to securely access all platform features.

Step 2: Create or Update Student Profile

After logging in, the student creates or updates their profile by providing academic and personal information such as department, batch, program, and other relevant details. Maintaining an updated profile helps the system provide better recommendations.

Step 3: Add Skills, Research Interests, and Career Goals

The student enriches their profile by specifying:

1. Technical skills
2. Research interests
3. Career aspirations or any related information according to their preference

Step 4: Explore Opportunities

From this point, the student can pursue one or more activities:

1. Browse Jobs & Internships: View available internship and job opportunities posted by alumni or administrators.
2. Apply for Opportunities: Submit applications for suitable positions.

Step 5: Search or Browse Recommended Mentors

The platform provides two ways to discover mentors:

1. Search Mentors: Manually search using filters such as department, industry, expertise, company, or skills.
2. Browse Recommended Mentors: View mentors suggested by the system based on the student's profile and interests.

Both options eventually allow the student to access detailed mentor

profiles.

Step 6: View Mentor Profile

The student reviews the mentor's profile, including:

1. Professional background
2. Skills and expertise
3. Current organization
4. Areas of mentorship
5. Experience
6. Previous mentoring activities

Step 7: Select Mentor

If satisfied with the mentor's profile, the student selects that mentor and proceeds with the mentorship request.

Step 8: Send Mentorship Request

The student sends a mentorship request, optionally including a personalized message explaining their learning objectives and expectations.

Step 9: Mentor Decision

The mentor reviews the request.

1. If Accepted: The mentorship relationship is established, and the student proceeds to the mentorship phase.
2. If Rejected: The student returns to the mentor search process to explore other mentors or recommendations.

Step 10: Chat with Mentor

Once the request is accepted, the student can communicate directly with the mentor using the platform's messaging system to discuss learning goals, ask questions, and receive guidance.

Step 11: Receive and Complete Tasks

The mentor assigns learning tasks, projects, or study materials. The student completes these activities according to the mentor's guidance.

Step 12: Track Learning Progress

The student monitors completed tasks and overall progress throughout the mentorship. This enables both the student and mentor to evaluate learning achievements and identify areas for improvement.

Step 13: Receive Mentor Feedback

After reviewing the student's performance, the mentor provides constructive feedback, suggestions for improvement, and recommendations for future learning.

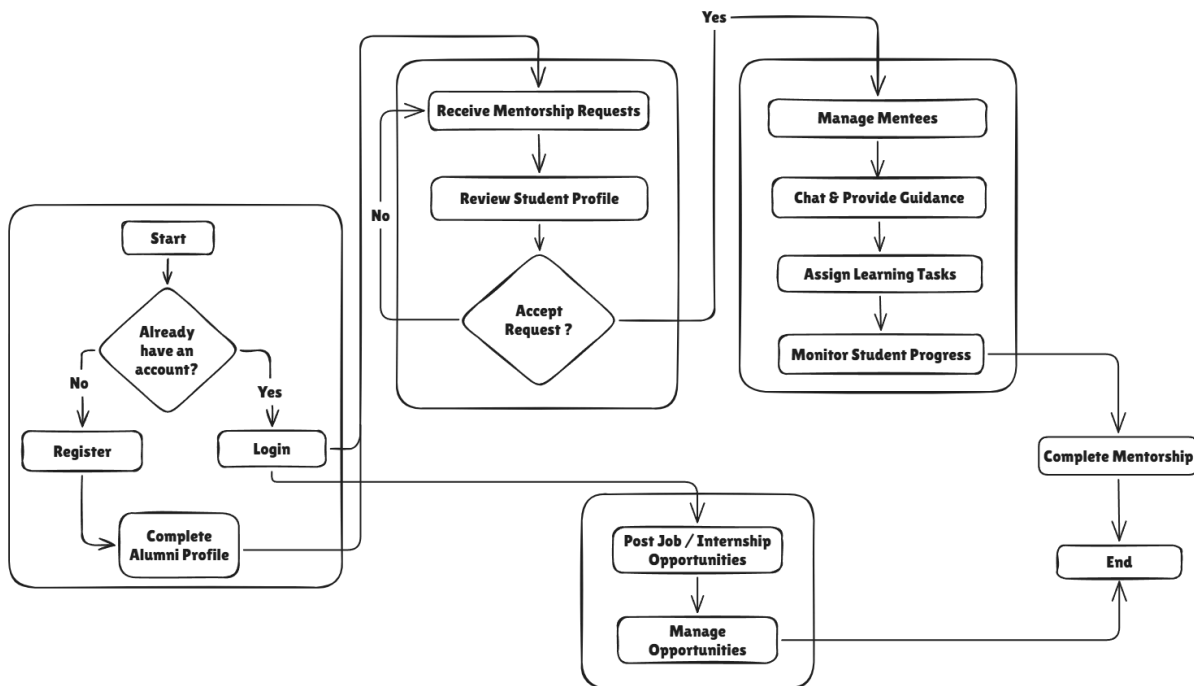
Step 14: Complete Mentorship

When all mentoring activities are completed, the mentorship is formally concluded. The student can then continue using the platform to:

1. Find additional mentors,
2. Apply for more internships or jobs,
3. Update their profile with new skills and achievements,
4. Participate in future mentoring opportunities.

Chapter 7

Alumni Workflow



Step 1: Account Registration/Login

The workflow begins when an alumnus opens the platform.

1. If the alumnus does not have an account, they proceed with the registration process by providing the required information (full name, email, password).
2. After successful registration, the alumnus completes their professional profile by adding details such as educational background, current company, job role, technical skills, areas of expertise.
3. If the alumnus already has an account, they simply log in using their credentials (email, password) to access the system.

Completing the profile ensures that students receive accurate mentor recommendations through the platform's attribute-based matching system.

Step 2: Receive Mentorship Requests

After logging in, the alumnus can receive mentorship requests submitted

by students.

For each request, the alumnus can:

1. View the student's profile
2. Review academic information, skills, career interests, and goals
3. Evaluate whether they are a suitable mentor for the student

This allows alumni to make decisions before accepting mentorship requests.

Step 3: Accept or Reject Requests

Once the student's profile has been reviewed, the alumnus decides whether to accept the mentorship request.

1. If the request is rejected, the workflow returns to the mentorship request list, allowing the alumnus to review other incoming requests.
2. If the request is accepted, a mentor–mentee relationship is established and the mentorship process begins.

This decision-making step ensures that mentorship connections are voluntary and based on compatibility.

Step 4: Manage Mentees

After accepting a request, the alumnus begins managing the assigned mentee.

The mentor can:

1. View active mentees
2. Track ongoing mentorship relationships
3. Organize mentoring sessions
4. Manage multiple students simultaneously

This stage serves as the central management point for all mentoring activities.

Step 5: Chat and Provide Guidance

The mentor communicates directly with the student through the platform's real-time messaging system.

Using this feature, the alumnus can:

1. Answer career-related questions
2. Share professional experiences
3. Provide interview preparation tips
4. Recommend learning resources
5. Offer internship and career advice

This continuous interaction strengthens the mentoring relationship and enables personalized guidance.

Step 6: Assign Learning Tasks

Based on the student's career goals and current skill level, the alumnus may assign learning tasks or recommend predefined career roadmaps.

These may include:

1. Learning specific technologies
2. Completing online courses
3. Building projects
4. Preparing for interviews
5. Following structured learning paths

This helps students develop the skills required for their desired career path.

Step 7: Monitor Student Progress

Throughout the mentorship, the alumnus monitors the student's progress.

1. The mentor can:
2. Review completed tasks
3. Track learning milestones
4. Evaluate improvement

5. Provide feedback and additional guidance

Monitoring progress ensures that students remain on track and receive continuous support.

Step 8: Post Career Opportunities

Apart from mentoring, alumni can contribute to the community by posting professional opportunities.

These opportunities include:

1. Internship openings
2. Job vacancies
3. Referral opportunities
4. Career advice

The alumnus can also update or manage previously posted opportunities, ensuring that students have access to current and relevant information.

Step 9: Complete Mentorship

Once the mentoring objectives have been achieved, the mentorship session is completed.

The mentor finalizes the mentoring relationship after the student has received sufficient guidance and support. The workflow then reaches its end, although the alumnus can continue mentoring new students or contribute further through career opportunities and community engagement.

Chapter 8

Conclusion

The ALUMNET platform serves as a vital link within the university community, effectively bridging the gap between current students and established alumni. By providing a structured environment for mentorship, networking, and career guidance, the platform allows students to navigate their professional journeys with confidence while enabling alumni to share their expertise and opportunities. Ultimately, ALUMNET establishes a system of professional development and collaborative engagement, ensuring that strong student–alumni relationships continue to drive career success and institutional excellence.